

Data Transfer of Master Data and Configurations MW 3.0/MW 3.1/MW 4.0pe

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1 Data transfer: master data and configurations

1.1 General

Version or release upgrades, the introduction of products or similar require master data and configurations to be transferred from one system (e.g. test or integration system) to another system (e.g. the live system).

But data should not be transferred all at once. The consultant and customer have to decide which objects they want to transfer and which not. The user can perform the data transfer with the aid of the lists outlined in this document.

1.2 Requirements for the data transfer

You must use the same product version for

- the system from which data is transferred and
- the system to which data is transferred.

1.3 Procedure for data transfer

A version or release upgrade is usually first carried out on a test system so that the changeover can be tested in advance. In the test system, make the configurations that may not be available until the new version. You want to transfer these configurations from the test system to the production system after a version or release upgrade.

It is necessary to be able to implement the transfer separately over time. The data transfer is therefore divided into two steps:

Exporting data

The *export script* unloads all relevant configurations from the source system into a shared directory.

Importing data

The *import script* selectively transfers this data to the target system. With each object, you are prompted to decide whether or not you want to transfer the object. If you want to transfer objects, the data existing in the corresponding table(s) of the target system is unloaded in a backup file and deleted from the tables. Then the previously exported data is loaded.

1.4 Restrictions for data transfer

The following restrictions apply if you transfer data using the scripts described in this document:



You can only transfer complete data of an objects. The data can consist of one or several tables. It is not possible to transfer single entries of an object (for example, you can only transfer a single order type "XY" or a single resource type).



If you want to transfer configurations from a test system as part of a version upgrade, you should note the following: all changes you made to the live system will get lost during the data transfer when you upgrade the version. To avoid this, you have to replicate all changes you made to the live system also in the test system to ensure these changes will be imported during the data transfer.



Not all configurations in the system are exported and transferred to a target system using the scripts described here. Configurations that are not transferred using these scripts must be transferred manually to the target system.



You cannot set up useful and extensive test systems for Access Control (ZKS) as this product group depends on the used hardware (access, terminals, and readers). Therefore, we do not recommend transferring master data.

1.5 Transferring data

The data transfer is performed with two scripts named <xxx>_export.scr and <xxx>_import.scr, where <xxx> is the name of the product group. The scripts are stored in the installation directory on the server. The example of the PZE script illustrates how to use these scripts.

1.5.1 Export configurations

Use the following commands in the server installation directory to start exporting configurations:

Windows:	sh pze_export.scr
UNIX:	pze_export.scr

The export process unloads all relevant configurations of the product group (here PZE) into the directory *hycfgunlpze*.

You can check if an error occurred during the export in the error log *hysql.err* of the directory "err" in the system directory. This file must not include current entries. The file would show the following entries if an error occurred:

```
03.02.2020 10:08:31.959 -942 SQL= select * from lohnarten_liste order by 1, 2;  
03.02.2020 10:08:31.959 -942 ERR=ORA-00942: table or view does not exist
```

1.5.2 Import configurations

In order to import configurations, copy the directory *hycfgunl* to the installation directory of the target system. Then start the script to import data:

Windows: sh pze_import.scr

UNIX: pze_import.scr

With each object, you are prompted to decide whether or not you want to transfer the object. Answer the prompt by choosing "Y" for "yes" or "N" for "no":

```

XXXXXXXX XXXXXX XXXXXX
X  X X  X  X  X
X  X  X  X
X  X  X  X  X
XXXXX  X  XXXX
X      X  X  X
X      X  X
X      X  X  X  X
XXXX  XXXXXX XXXXXX

XXXXX  XX  XX XXXXXX  XXX  XXXXXX XXXXXX
X  X  X  X  X  X  X  X  X  X  X  X
X  XX XX  X  X  X  X  X  X  X
X  XX XX  X  X  X  X  X  X  X
X  X  X  XXXXX  X  X  XXXXX  X
X  X  X  X  X  X  X  X  X
X  X  X  X  X  X  X  X  X
X  X  X  X  X  X  X  X  X
XXXXX  XXX XXX XXXX  XXX  XXX  XX  XXX

-----
*** Loading PZE/PZW configurations
-----

*** Account limits (Kontogrenzen) (Y/N)
-----
N
-----

*** PZE/PZW Public holidays (Feiertage) (Y/N)
-----
Y
-----

19.11.2019 10:13:36 PROCESSING STDIN...

SQL> xunload to ./hycfgunl/pze/pze_feiertage.sav_unl select * from pze_feiertage order by 1, 2;
OK. NR OF ROWS 70.

SQL>
19.11.2019 10:13:36 PROCESSING STDIN...

SQL> delete from pze_feiertage;
OK. NR OF ROWS 70.

SQL>

19.11.2019 10:13:36 PROCESSING STDIN...

SQL> load from ./hycfgunl/pze/pze_feiertage.unl insert into pze_feiertage;
OK. NR OF ROWS 70.

SQL>
-----
...

```

You can check if an error occurred during the import in the error log *hysql.err* of the directory "err" in the system directory. This file must not include current entries.

2 Checklists for the data transfer

This chapter deals with checklists you have to complete and return to Implementation Services.

The checklists of the single product groups sometimes include general objects (e.g. basic settings, users, function authorizations, responsibility areas). You only have to check these objects in one of the checklists if you transfer configurations for multiple product groups. It goes without saying that the data transfer will still be correct, even if you select these objects in multiple lists. Only the backup copy you made before will be overwritten by the transferred data.

Special notes apply to specific data objects. These objects are highlighted in the table with the below-described icons:

Further information

There are additional information about the object, e.g. which data is included or which configuration applications are affected.

Please note:

Additional notes apply to the object. Such notes indicate, for example, that transferring MLE variants affects the entire system and not only the current product group.

The notes are added below the checklists.

2.1 Checklist for the PZE/PZW/PEP data transfer

If PZE/PZW configurations should be transferred, the following checklist must be completed by the consultant and/or customer.

Script files:

pze_export.scr

pze_import.scr

Object	Transfer data		
	Yes	No	
Account limits			
PZE/PZW public holidays			
Periods for overtime calculation			
Settlement periods			
Breaks depending on working time			
Clocking authorizations			
Cost centers			
Working time day types			
Working time models			
Shift rhythm models			
Payment day types Payment rules			
Payment models			
Payment depending on shift type			
Wage types Update accounts Additional allowance rules Wage types relations			
Control of absences			
Control of labor time calculation			
Absence reasons			
Wage type groups Configuration of wage type statistics			
Time sheet configuration			
Configuration of message listings			

Object	Transfer data		
	Yes	No	
Configuration of accounts			
Configuration of leave entitlement			
Configuration of HR master data fields			
Configuration of data management for PZE/PZW tables			
Scheduler jobs for PZE/PZW			
Customization the PZW interface to the payroll system (Customization for the PZW interface used in the payroll accounting)			
Basic settings of Escalation Management			
Function groups [Escalation Management] (Function groups [Escalations Management])			
Escalation configuration for PZE/PZW (Escalation configuration for PZE/PZW)			
Qualifications			
Staff qualifications			
Workforce requirements of workplaces			
Terminal configuration for PZE terminals			
Terminal groups			
PZE 8.2: Configuration of terminal information			
User administration Users Function authorizations Responsibility areas			



Wage types

Wage types also affect the product group Premium & Incentive Wages (LLE).



Time sheet configuration

Once you have transferred the **time sheet configuration**, also copy the corresponding reports to the target system. These are the files PersonalTimeSheet_*.lul in the system directory <x>\custom\reports located on the server.

 Configuration of message listings

The message listings also affect the product group Premium & Incentive Wages (LLE).

 Configuration of HR master data fields

Not only does Personnel Time Management use the configurable HR master fields, also other product groups require these fields to configure software modifications. You have to make sure that **all** HR master fields have been configured in the source system.

 Configuration of data management for PZE/PZW tables

Configurations for the product groups PZE/PZW are transferred.

 Scheduler jobs for PZE/PZW

Configurations for the product key PZE are transferred.

 Customizing the PZW interface to the payroll system

The INI configuration "HYD-LUG" is transferred. No further INI configurations are adopted.

 Basic settings of Escalation Management

These objects affect other product groups!

 Function groups [Escalation Management]

These objects affect other product groups!

 Escalation configuration for PZE/PZW

The data transfer is restricted to the PZE/PZW escalations.

 Terminal configuration for PZE terminals

All terminals where PZE is enabled are transferred. These terminals are transferred completely, i.e. also the settings for other product groups!

 Terminal groups

These objects affect other product groups!

 User administration

These objects affect other product groups!

2.2 Checklist for the LLE data transfer

If LLE configurations should be transferred, the following checklist must be completed by the consultant and/or customer.

Script files:

lle_export.scr

lle_import.scr

Object	Transfer data		
	Yes	No	
Basic settings for incentive wages (LLE)			
Bonus reasons			
Wage type determination			
Premium groups			
Assignment of premium groups			
Assignment of premium areas			
Premium accounts			
Configuration of data management for LLE tables			
Scheduler jobs for LLE			
User exits for LLE			
Wage types			
Configuration of message listings No. 900 to 950 for LLE			
Configuration of HR master data fields			
User administration Users Function authorizations Responsibility areas			



Premium accounts

Premium accounts are user field configurations. Only user field configurations relating to premium accounts are transferred. Other user field configurations are not affected.

 Configuration of data management for LLE tables

Configuration for the product group LLE are transferred.

 Scheduler jobs for LLE

Configuration for the product key LLE are transferred.

 User exits for LLE

The user exits of the LLE product group are transferred. These are all HYDRA script user exits starting with "hyl", "lrck", "lpb", "lpv", "lsl", "lsz" or "lsv". User exits having other names will not be transferred and remain unchanged in the target system.

 Wage types

Wage types also affect Personnel Time Management (PZW). Only wage types are transferred. The following objects are excluded: "Update accounts", "Additional allowances rules" and "Wage type relations".

 Configuration of HR master data fields

Not only does Personnel Time Management use the configurable HR master fields, also other product groups require these fields to configure software modifications. You have to make sure that **all** HR master fields have been configured in the source system.

 User administration

These objects affect other product groups!

2.3 Checklist for the data transfer Workplace/machine or resources

Configurations for the product groups PZE/PZW are transferred.

Script files:

mde_export.scr

mde_import.scr



In order to import Workplace/machine (resources of type MNR) to the target system, NO operations, NO persons, NO resources, NO batches must be logged in at the time of import.



To import resources (type <> MNR) into the target system, NO resources must be logged on at the time of the import.

Object	Transfer data		
	Yes	No	
<i>MDE master data</i>			
Shift models (BDE)			
Year model			
Day types			
Days off			
RPA/status class/status texts			
RPA			
Status classes			
Status texts			

Object	Transfer data		
	Yes	No	
Workplace/machine configuration Workplace configuration Resource attributes Status assignment Resource status assignment Counter configuration Process parameters Operator positions Premium indicators Compensation rules (EMG compensation rules; only for workplaces/machines) Resource status types Resource status text Parallel resource status Terminal assignment Line assignment Assignment of DNC family to machine Groups Group assignment (only workplaces/machines) Activity/Maintenance/Calibration calendar (only workplaces/machines) Production variants (only for entries in the workplace field) The following data is only transferd if MDE and WRM master data is transferred. Resource list (assignment of data that have a higher level resource of resource type MNR) Assignment of counter to machine			

Object	Transfer data		
	Yes	No	
WRM master data			
Resource attributes Resource status assignment			



MDE master data

○

○



Shop floor monitor (graphic machinery)

2.4

Script files:

mf_export.scr

mf_import.scr

Object	Transfer data		
	Yes	No	
<i>General customization data</i>			
Units Units Unit conversion Formula management			
User field configuration			
HYD-INI configuration			
Enhanced object configuration			
Number ranges			
Archiving configuration Data management Reload Manager			
Path configuration			
Scheduler configuration (not PZE, ZKS and LLE)			
Logging configuration Logging keys			
Escalation configuration			
<i>Basic settings</i>			
General setup settings			

Object	Transfer data		
	Yes	No	
<i>Order customization data</i>			
Order customization (order configurations) Order types Order status texts Order status assignment Processing codes			
Grund			
Order and operation statuses, customized data sources			
<i>MLE configuration</i>			
MLE variants MLE basic configuration MLE segment configuration MLE field configuration MLE formulas / conditions Formula management			
Distribution model Logical systems Distribution model			
SAP Setup SAP order sequencing SAP upload SAP activity types			
Configuration of label printing			
User administration Users Function authorizations Responsibility areas			
Terminal BDE configurations			
Terminal groups			
<i>HLS Customization data</i>			
HLS sorting rules			

Object	Transfer data		
	Yes	No	
HLS planning profiles			
HLS setup			
<i>MPL customization data</i>			
Transport units Transport units			
Material buffers Material buffers			
MPL setup			
<i>Dynamic dialogs</i>			
Dynamic dialogs Only user 0 dialogs (user 0 dialogs) (CTWIN+AIP) All dialogs			



User field configuration

All user fields including user field keys and user field types are transferred. Even those configurations not deriving from PDV are transferred.



Escalation configuration



SAP Setup

These objects affect other product groups!



Terminal groups

These objects affect other product groups!

2.5 Checklist for the CAQ data transfer

Script files:

caq_export.scr

caq_import.scr



MDE master data or transfer of WRM customization data

Object	Transfer data		
	Yes	No	
General customization data			
Units Units Unit conversion Formula management			
User field configuration			
INI configuration			
Enhanced object configuration			
Number ranges			
Archiving of data management configurations Data management objects			
Path configuration			
Scheduler configuration			
Logging configuration			
Signature types Signature matrix			
Escalation configuration Escalation configuration Function groups			
User administration			

Object	Transfer data		
	Yes	No	
User administration Users Function authorizations Responsibility areas Responsibility profiles Assignment: User ↔ Responsibility area Function profiles Password policies			
Terminal CAQ configurations Note: The settings made in the "QM functions" tab are not transferred. You have to replicate these settings manually in the live system. Usually, this only affects the option "Inspector identification required before opening inspection dialog".	Not applicable (see the note on the left-hand side)		
Order customization data			
Order customization (order configurations) Order types Order status assignment Processing codes Order status texts			
MLE configuration			
MLE variants MLE basic configuration MLE segment configuration MLE field configuration MLE formulas / conditions Formula management MES types MES conversion functions Content of the table "caq_quails_vormerk"			
Distribution model Logical systems Distribution model			
SAP Setup SAP order sequencing SAP upload SAP activity types Basic settings (order and operation number lengths, etc.) SAP basic settings (SAP order and operation number lengths, etc.)			
Data specific to CAQ			
Options			
Status types			
Statuses			
Areas			

Object	Transfer data		
	Yes	No	
Forms			
Assignment of order types to CAQ areas			
Catalog			
Dynamic modification master data Inspection severities (including all entries) Transitional definitions including entries Dynamic modification norms including AQL, inspection levels, methods, sampling plans			
QM activity, machine status change			
Variable workplaces			
<i>Dynamic dialogs</i>			
Dynamic dialogs Only user 0 dialogs (user 0 dialogs) (CTWIN+AIP) All dialogs			



Units

All units are transferred. Even if these units are not used in CAQ applications are transferred.



User field configuration

All user fields including user field keys and user field types are transferred. Even those configurations not deriving from CAQ are transferred.



INI configuration

All INI configurations available in the system are transferred.



Enhanced object configuration

All entries of the enhanced object configuration are transferred. Even those configurations not deriving from CAQ applications are transferred.



Number ranges

All number ranges available in the system are transferred. Even those number ranges not affecting the CAQ product group are transferred. The order type KAL and its corresponding number range are relevant to CAQ PMV.

Archiving of data management configurations

All archiving configurations from the data management table of the product groups "CAQ" and "QMS" are transferred for all objects. The corresponding object configurations of data management are transferred in addition to all transferred entries.

Scheduler configuration

Only scheduler configurations with a product key **not** starting with PZE, ZKS or LLE are transferred.

Path configuration

All path configurations available in the system are transferred. Even those configurations not deriving from CAQ are transferred. CAQ requires, for example, the path EFORMCUS where customized forms are stored.

Logging configuration

All logging configurations available in the system are transferred. Even those configurations not relating to CAQ applications are transferred.

Signature types (authorization types)

All signature types and entries of the signature matrix available in the system are transferred. Even those configurations not relating to CAQ applications are transferred. --> Currently, not used in CAQ.

Escalation configuration

All escalation configurations and function groups available in the system are transferred. The escalation setup is also exported and imported. Therefore, also configurations not relating to CAQ applications are transferred.

User administration

All users available in the system's user administration are transferred. All existing function authorizations, responsibility profiles, responsibility areas, the users' assignments to responsibility areas and functions profiles are transferred. Consequently, even those users, function authorizations and responsibility profiles, etc. not connected to CAQ users will also be transferred.

Order customization (order configurations)

All order types, assignments of order statuses, processing codes and order status texts are transferred. Even those configurations not relating to CAQ applications are transferred.



MLE variants

The following is transferred: all MLE basic, segment and field configurations, MLE formulas and contents of formula management. MES types and MES conversion functions are also transferred. In contrast to the MF data transfer, the contents of the table *caq_quails_vormerk* are also transferred. Consequently, configurations/database contents not relating to CAQ applications are also transferred.



Distribution model

All distribution models and logical system entries existing in the system are transferred. Even those data records not relating to CAQ applications are transferred.



SAP Setup

All SAP setup settings are transferred including: "SAP order sequencing", "SAP uploads", "SAP activity types" and the basic parameter settings for the system and for the communication with SAP (e.g. order and operation number lengths, etc.). Even those settings not relating to CAQ applications are transferred.



Dynamic dialogs

All dynamic dialogs pertaining to user 0 and all other dialogs relating to CTWIN and AIP are transferred. Additionally, all fields and function keys relating to these dynamic dialogs are transferred. Workflow configurations are also transferred.



CAQ options

All CAQ system options are transferred. CAQ system options manage fundamental functions and processing of CAQ and PDV.



Status types

All CAQ status types are transferred. They are the basis for statuses. You can use CAQ statuses to adjust CAQ functions, to a certain extent, to your business processes.



Status

All CAQ statuses are transferred. You can use CAQ statuses to adjust CAQ functions, to a certain extent, to your business processes.



Areas

All CAQ areas are transferred. Areas define the functional fields and substructures CAQ is divided into. Example: area F for production, area E for goods receipt.



Forms

All CAQ forms are transferred. The forms include standard and customized Word reports.

**Assignment of order types to CAQ areas**

Contents of the table *ade_aart_cbereich* are transferred. These contents are only relevant to CAQ. This table specifies, for example, if the logon of an OP triggers the search for an active inspection plan.

**Catalogs**

All SAP-QM catalogs are transferred. If the system is not used with SAP-QM, this table also includes the usage decisions for inspection points. These decisions are required for inspections relating to inspection points.

**Dynamic modification master data**

Table contents relating to inspection severities, transitional definitions and dynamic modification norms are transferred. Their corresponding entries are also transferred (inspection severities of an inspection severity definition, transitional definitions of a transitional definition, sampling plans of dynamic modification norms). Contents of the tables relating to AQL values, inspection methods and inspection levels are also transferred.

**Dynamic modification master data**

Table contents relating to inspection severities, transitional definitions and dynamic modification norms are transferred. Their corresponding entries are also transferred (inspection severities of an inspection severity definition, transitional definitions of a transitional definition, sampling plans of dynamic modification norms). Contents of the tables relating to AQL values, inspection methods and inspection levels are also transferred.

**QM activity, machine status change**

These entries are only relevant if you use the system as a QM subsystem, e.g. in connection with SAP-QM. These entries specify with respect to the workstation, whether a status change of the machine where an inspection step is logged on, is to trigger an activity (due inspection). This activity relates to the inspection step that is currently logged on. The generation of an inspection point could be such an activity.

**Variable workplaces**

These entries are only relevant if you use the system as a QM subsystem, e.g. in connection with SAP-QM. Although inspection planning is performed in a higher-level system (e.g. SAP-QM), these entries address inspection planners. The entries specify which inspections (QM operations/processes) are performed at which workstations.

2.6 Checklist for the PDV data transfer

If PDV configurations should be transferred, the following checklist must be completed by the consultant and/or customer.

Script files:

pdv_export.scr

pdv_import.scr



MDE master data or transfer of WRM customization data

If you transfer data from the MDE master data or MDE/WRM customizing data, the scripts described in 2.3 Checklist for the data transfer of workplaces/machines or resources are product independent and can be executed additionally.

Object	Transfer data		
	Yes	No	
General customization data			
Units Units Unit conversion Formula management			
User field configuration			
INI configuration			
Number ranges			
Terminal configuration			
Archiving of data management configurations Data management objects			
Path configuration			
Scheduler configuration			
Logging configuration Logging keys			
Signature types Signature matrix			
Escalation configuration Escalation configuration Function groups			

Object	Transfer data		
	Yes	No	
User administration			
User administration Users Function authorizations Responsibility areas Responsibility profiles Assignment: User ↔ Responsibility area Function profiles Password policies			
Data specific to CAQ			
Options			
Status types			
Statuses			
Areas			
Assignment of order types to CAQ areas			
TNT configurations			
TNT configurations TNT header configuration TNT column configuration			
Data specific to PDV			
Data specific to PDV			
PDV settings			
PDV settings			



Units

All units are transferred. Even those units not used in PDV applications are transferred.



User field configuration

All user fields including user field keys and user field types are transferred. Even those configurations not deriving from PDV are transferred.



INI configuration

All INI configurations available in the system are transferred.



Number ranges

All number ranges available in the system are transferred. Even those number ranges not affecting the PDV product group are transferred.



Archiving of data management configurations

All archiving configurations from the data management table of the product groups "CAQ" (for the objects "FEP", "A_FEP") and "PDV72" are transferred for all objects. The corresponding object configurations of data management are transferred in addition to all transferred entries.



Scheduler configuration

Only scheduler configurations with a product key **not** starting with PZE, ZKS or LLE are transferred.



Path configuration

All path configurations available in the system are transferred. Even those configurations not deriving from PDV are transferred.



Logging configuration

All logging configurations available in the system are transferred. Even those configurations not relating to PDV applications are transferred.



Signature types (authorization types)

All signature types and entries of the signature matrix available in the system are transferred. Even those configurations not relating to PDV applications are transferred.



Escalation configuration

All escalation configurations and function groups available in the system are transferred. The escalation setup is also exported and imported. Therefore, also configurations not relating to PDV applications are transferred.



TNT configurations

Apart from the tables, all required configuration files are copied:

- tntcfg_2<terminal number PCC>.xml
- sqlldr_<table-id>.tpl

**User administration**

All users available in the system's user administration are transferred. All existing function authorizations, responsibility profiles, responsibility areas, the users' assignments to responsibility areas and functions profiles are transferred. Consequently, even those users, function authorizations and responsibility profiles, etc. not connected to PDV users will also be transferred.

**CAQ options**

All CAQ system options are transferred. CAQ system options manage fundamental functions and processing of CAQ and/or PDV.

**Status types**

All CAQ status types are transferred. They are the basis for statuses. You can use CAQ statuses to adjust CAQ functions, to a certain extent, to your business processes.

**Status**

All CAQ statuses are transferred. You can use CAQ statuses to adjust CAQ functions, to a certain extent, to your business processes.

**Areas**

All CAQ areas are transferred. Areas define the functional fields and substructures CAQ is divided into. Example: area F for production, area E for goods receipt, PDV and PDAG for PDV.

**Assignment of order types to CAQ areas**

Contents of the table *ade_aart_cbereich* are transferred. These contents are only relevant to CAQ.